

ML Lab Week 14: CNN Image Classification

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1. Introduction

This lab is on building a Convolutional Neural Network (CNN) to classify hand gesture images into three categories: rock, paper, and scissors. The goal was to train a deep learning model that can automatically recognize and distinguish between these hand gestures. This involved loading and preprocessing image data, designing a CNN architecture, training the model, and evaluating its performance on unseen test images.

2. Model Architecture

The CNN model has two main parts: a feature extraction section and a classification section.

Convolutional Blocks (Feature Extraction):

The model uses three convolutional blocks to learn features from images. Each block has a Conv2D layer, ReLU activation, and MaxPooling layer. The first block converts the 3-channel RGB image into 16 feature channels using a 3×3 kernel with padding. The second block increases this to 32 channels, and the third block expands it to 64 channels. After each convolutional layer, MaxPooling reduces the image size by half, making the model more efficient while keeping important features.

Fully-Connected Classifier:

After the convolutional blocks, the features are flattened into a single vector. This vector is passed through a dense layer with 256 neurons and ReLU activation. Dropout (30%) is applied to prevent overfitting. Finally, an output layer with 3 neurons produces the prediction for rock, paper, or scissors.

3. Training and Performance

Hyperparameters:

- Optimizer: **Adam optimizer** with a learning rate of 0.001
- Loss Function: Cross Entropy Loss (standard for multi-class classification)
- Batch Size: 32 images per batch
- Epochs: 10 training cycles
- Train-Test Split: 80% training, 20% testing

Final Test Accuracy: 98.86%

The training loss decreased steadily from 0.68 in epoch 1 to near zero by epoch 10, showing the model learned the patterns effectively.

4. Conclusion and Analysis

The model performed very well, achieving 98.86% accuracy on the test set. This high accuracy shows that the CNN successfully learned to distinguish between rock, paper, and scissors gestures. The training was smooth with consistent loss reduction, indicating no major issues like overfitting or underfitting.

Challenges:

The main challenge was ensuring proper data preprocessing and setting up the correct image transformations. Getting the tensor dimensions right for the fully-connected layers required careful calculation based on the pooling operations.

Potential Improvements:

1. Data Augmentation: Adding random rotations, flips, and brightness changes during training could make the model more robust to different hand positions and lighting conditions.
2. Deeper Architecture: Adding more convolutional layers or using techniques like batch normalization could potentially capture even more complex patterns, though with 98.86% accuracy, there's minimal room for improvement.